

INPUT

M METAMODEL

DCL

CSL

FSL

Predicates C1–C9

Traces + format[1]

Cross-layer graph predicates

S SCENARIO

Housing subsidy (CSOK)

3 credentials

3 issuers

Cross-credential deps.

Value constraints

G GOVERNANCE CONSTRAINTS

eIDAS ARF

GDPR 5(1)(c)

W3C VCDM 2.0

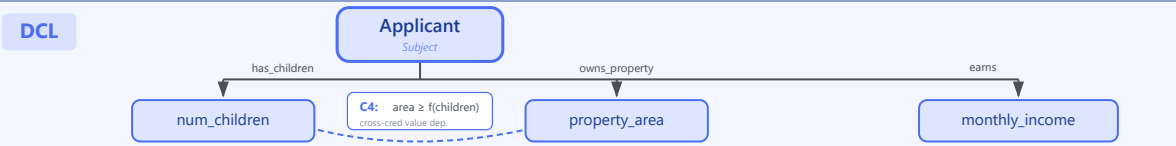
Format mandates

Privacy reqs.

Data model conformance

MODEL

DCL



C1: entity alignment
all subjects → same Applicant

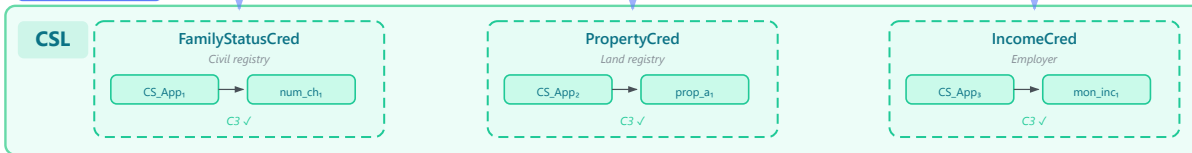
C2: trace consistency
each Claim traces to one Prop

trace[1]

trace[1]

trace[1]

CSL



C5: eIDAS mandate
EU wallets → SD-JWT-VC

format[1]

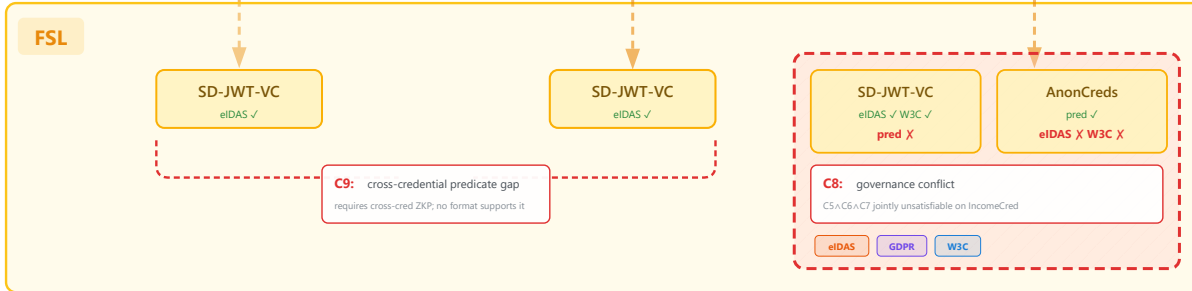
C6: pred proof req.
GDPR → income needs ZKP

format[1]

C7: VCDM conform.
W3C → AnonCreds fails

format[1]

FSL



C9: cross-credential predicate gap
requires cross-cred ZKP; no format supports it

C8: governance conflict
C5 ∧ C6 ∧ C7 jointly unsatisfiable on IncomeCred

RESULT

1 VERIFY

SAT

design is consistent

UNSAT

constraints conflict

2 IDENTIFY ERRORS

▪ **governance_conflict**

(IncomeCred)

▪ **cross_cred_pred_gap**

(area, children)

3 GENERATE

Relax C6 (GDPR) → 2 configurations



✓ all SD-JWT-VC



✓ Prop → mdoc